

Lecture 2

Last Time:

"Given the truth of some statements, what other statements can we deduce."

proposition $\begin{matrix} \swarrow T \\ \searrow F \end{matrix}$
 $\neg, \vee, \wedge, \rightarrow$ ops

Compound proposition. $(p \rightarrow q) \vee (q \rightarrow r) \equiv T$
tautology.

logic puzzle: If $p \leftrightarrow (p \leftrightarrow q)$ is T then q is T.

I Predicate logic, Quantifiers II Rules of Inference III ~~Syllabus~~

Motivation: "Every integer is even or odd."

Attempt: $(0 \text{ is even or odd}) \wedge (1 \text{ is even or odd}) \wedge (2 \text{ is even or odd}) \dots$

Defn: A predicate is a statement containing one or more variables from a domain, which becomes a proposition when each variable is instantiated.

ex 1: $P(x) = "x + 2 = 5."$ Domain
 x integer

$$\text{even}(x) = "x \text{ is even}."$$

$$\text{odd}(x) = "x \text{ is odd}."$$

x integer
"

Domain must be specified, explicitly or implicitly.

$$R(x, y) = "x < y." \quad x, y \text{ real numbers}$$

★ Quantifiers

Defn: (Universal)

$\forall x P(x) := " \text{For every } x \text{ in the domain, } P(x). "$

define LHS as RHS.

ex2:

$$\forall x (\text{even}(x) \vee \text{odd}(x)) \quad x \text{ integer.} \quad T$$

$$\forall x, y \quad \{ \quad \forall x \forall y (R(x, y)) \quad x, y \text{ real.} \quad F$$

$$\forall y \forall x \quad R(x, y)$$

Defn: (Existential)

$\exists x P(x)$ = "There exists an x in the domain such that $P(x)$."

ex3: $\exists x (x^2=2)$, x real. T , $x=\sqrt{2}$

$\exists x (x^2=2)$, x integer. F

$\exists x, y$ $\{ \exists x \exists y (x^2=2 \wedge y^2=2 \wedge x \neq y), x, y$ real T

Terminology: A variable appearing in a quantifier is bound.

Otherwise it is free.

A statement in which all variables are bound is a proposition.

Q7) " $0 \cdot x = 0$ " should be: $\forall x (x \cdot 0 = 0)$, x integer.
free variable

Remarks: The same stmt can be written using logic notation
• or English (not unique).

- If the domain is finite $x_1, \dots, x_n$

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n).$$

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n).$$

- Can use $\rightarrow$ to specify the domain:

$$\text{ex2} \equiv \forall x (x \text{ is an integer} \rightarrow \text{even}(x) \vee \text{odd}(x))$$

$$\text{ex3} \equiv \exists x (x \text{ is real} \wedge x^2 = 2).$$

[cf $\exists x (x \text{ is real} \rightarrow x^2 = 2)??$
vacuously satisfied
by x not real.]

Negation of $\forall, \exists$:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

ex4:

$\neg$ (Every integer is prime.)

Negation.

There exists an integer which is not prime.

logically equiv:
same truth value for
all predicates & domains

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

cf:

"De Morgan's Laws":

$$\neg(p \vee q) = \neg p \wedge \neg q$$

$$\neg(p \wedge q) = \neg p \vee \neg q$$

propositional

Nesting of $\forall, \exists$:

ex 5: We can write expressions like:

$$\forall x \underbrace{(\exists y (x < y))}_{\text{Predicate in } x} \quad x, y \text{ integers } \mathbf{T}$$

"For every integer x , there exists an integer y s.t. $x < y$."

Swap: $\exists y \forall x (x < y) \quad \text{integers.}$

"There exists an integer y s.t. for every x , $x < y$." $\mathbf{F}$

~~*~~ In general,

$$\forall x \exists y R(x, y) \neq \exists y \forall x R(x, y)$$

Pol1: $\neg (\forall x \exists y (P(x) \wedge Q(y)))$

$\equiv$
 $\neg \forall \equiv \exists \neg$

$\exists x \neg (\exists y P(x) \wedge Q(y))$

$\equiv$
 $\neg \exists \equiv \forall \neg$

$\exists x \forall y \neg (P(x) \wedge Q(y))$

$\equiv$
de Morgan

$\exists x \forall y (\neg P(x) \vee \neg Q(y))$

II Rules of Inference i.e. useful tautologies used to deduce new propositions from old.

- Can use $\equiv$ to infer truth values.

- For propositions:

$$\begin{array}{l} \textcircled{1} \quad \begin{array}{l} P \rightarrow q \\ P \end{array} \quad \left. \vphantom{\begin{array}{l} P \rightarrow q \\ P \end{array}} \right\} \begin{array}{l} \text{premises,} \\ \text{assume T} \end{array} \\ \hline \therefore q \quad \left. \vphantom{\therefore q} \right\} \text{conclusion.} \end{array}$$

$$\begin{array}{l} \textcircled{2} \quad P \rightarrow q \\ \neg q \\ \hline \therefore \neg P \end{array}$$

$$\begin{array}{l} \textcircled{3} \quad P \rightarrow q \\ q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array}$$

If it is raining, the dog is wet.
 P It is raining

 $\therefore$ The dog is wet. q

If it is raining, the dog is wet.
The dog is not wet
 $\therefore$ It is not raining.

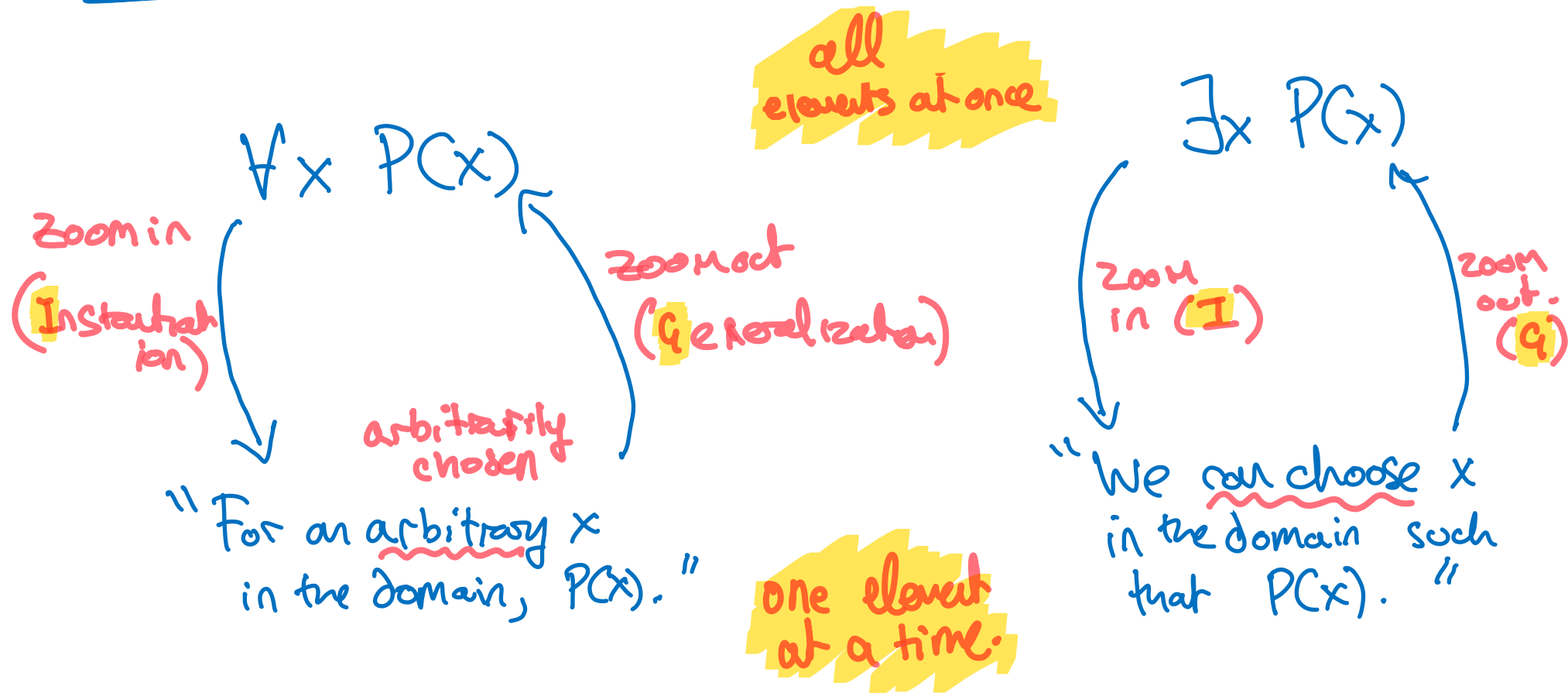
Logical Fallacy:

$$\frac{P \rightarrow Q}{\neg P} \quad \frac{\quad}{\therefore \neg Q}$$

Not always valid.

• ROI for $\forall, \exists$

→ "therefore!"



ex 6:

$\forall x (x \text{ is mortal.})$ people in room (I)

 $\therefore$ Neel is mortal.
⋮

ex 7:

$\exists x (x \text{ is wearing a white t-shirt})$ (I)

 $\therefore$ It is possible to choose x
with a white t-shirt.

Nikhil is wearing a white t-shirt. (9)

 $\therefore \exists x (x \text{ is wearing a white t-shirt}).$

ex 8:

Pick an arbitrary person x .
 x has red hair.

← regardless of x .

$\therefore \forall x$ (x has no hair). (9)

Addendum to Lecture 2

Two definitions needed for HW1:

Defn: The contrapositive of $p \rightarrow q$ is $\boxed{\neg q \rightarrow \neg p}$.

exercise: Check that $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

Defn: An argument is a sequence of propositions, each of which is a premise (i.e. an assumption) or a conclusion. It is valid if each conclusion follows from the previous propositions via rules of inference.